



D100 – FUNDAMENTALS OF DATA SCIENCE

COURSE OUTLINE 2025-26

LECTURERS: Dr. Adrian Ochs & Dr. Christian Rörig

OFFICE HOURS: By appointment, please email Adrian Ochs at acro2@cam.ac.uk and Christian Rörig at cmr61@cam.ac.uk

FORMAT OF COURSE:

Teaching Hours Overview:

- 18 hours of Lectures taught in Michaelmas Term; comprising of 9 lectures (2 hours each).
- Classes where assigned.
- Students will work in groups for class assignments.

COURSE OVERVIEW:

Course Description:

This course offers a practical introduction to the fundamental concepts and tools of data science. You'll learn to manage end-to-end data science projects, from data acquisition and preprocessing to analysis, visualisation, and statistical modelling. The curriculum is designed to build the essential skills for data science interviews and careers, covering not just data science methods but also valuable software engineering practices.

Specific Topics Covered:

The course follows the lifecycle of a typical data science project:

1. Lifecycle of a data science project
 - Introduction to typical stages of a data science project
2. Project setup phase
 - Initial project scoping and trade-offs
 - Setting up a data science repository
 - Setup of pre-commit hooks, environment.yaml, .gitignore, etc.
3. Data retrieval and storage
 - Data types and formats
 - Loading data files (e.g., CSV, Parquet)
 - Working with SQL databases
4. Data wrangling and exploration
 - Hands-on data exploration sessions using Python's pandas and polars libraries
 - Coping with messy real-world data (e.g. missing values, outliers)
5. Data visualisation
 - Best practices for effective visualisations
 - Utilising libraries such as matplotlib, seaborn, plotly, and altair
6. Feature Engineering
 - Feature engineering for numeric and categorical features
 - Dimension reduction techniques
 - FE pipelines using scikit-learn
7. Statistical Modelling
 - Modelling strategy
 - Cross-Validation
 - Feature selection
 - Hyperparameter tuning
 - Advanced modelling techniques (e.g. model composition, custom loss function)
8. Model evaluation and interpretability
 - Evaluation metrics
 - Interpreting models using model-agnostic methods
9. Model tracking and reproducibility
 - Using MLflow for experiment tracking
 - Model conversion to ONNX format

Course Aims and Objectives:

By the end of this course, you'll be able to:

- **End-to-End Project Execution:** manage a complete data science project—from initial scoping and data wrangling to modelling, evaluation, and considering production needs.
- **Technical Proficiency:** proficiently use core Python libraries like pandas and scikit-learn to explore data, build feature engineering pipelines, and implement statistical models.
- **Best Practices:** implement and review good coding practices to ensure your work is reproducible and maintains a high standard of quality.
- **Teamwork:** collaborate effectively on a shared codebase using industry-standard tools like Git and GitHub, including pull requests and code reviews.
- **Portfolio Building:** By the end of this course, you will have developed a significant data science project that can serve as a foundational piece for your professional portfolio on GitHub.

Lecture Notes:

Available via Moodle and the course's GitHub repository [links will follow].

Computing Environment:

Your laptop with:

- VS Code as IDE
- Conda for package management
- Python as the primary programming language
- Git and GitHub for version control and collaboration

RESOURCES AND READING MATERIALS:

This is an applied course that incorporates practical insights and workflows from the private sector. We don't follow a single textbook; instead, we will point you to key articles, documentation, and other relevant materials during the lectures.

If you are looking for one main book to help you prepare, we recommend An Introduction to Statistical Learning by James et al. (<https://www.statlearning.com/>).

Resource:	Description:
https://www.statlearning.com/	An Introduction to Statistical Learning with Applications in Python

ASSESSMENT AND EXAMINATION STRUCTURE:

Please refer to the Form and Conduct Notice 2025 – 2026 which will be released in Michaelmas Term.